

VIII. Least Upper Bound

Def. Let $A \subseteq \mathbb{R}$, then A is bounded above if $\exists \alpha \in \mathbb{R}$ s.t. $\forall x \in A, x \leq \alpha$
 $\Rightarrow \alpha$ is an upper bound for A

Ex. ① $A_1 := \{-x^2 : x \in \mathbb{R}\}$ is bounded above. $\nearrow 0 = \sup A_1 = \max A_1$. 100 (or 0) is an upper bound for the set.

② $A_2 := \{-\frac{1}{x} : x > 0\}$ is bounded above by 0 ($0 \notin A_2$). $0 = \sup A_2$
 $\max \text{ DNE}$

③ $A = \mathbb{R}$ is not bounded above

Rmk. A function f is bounded above $[a, b]$ if the set $\{f(x) : x \in [a, b]\}$ is bounded above.

Def. If $A \subseteq \mathbb{R}$, $\alpha \in \mathbb{R}$ is the least upper bound for A if

① α is an upper bound of A

② If β is also an upper bound for A , then $\alpha \leq \beta$

Use the notation, $\alpha = \sup A$.

(for greatest lower bound, use the notation $\inf A$).

Lemma. LUB of a set (if it exists) is unique.

Pf. Let α, β be LUBs of a set A

$$\alpha \text{ LUB} + \beta \text{ UB} \Rightarrow \alpha \leq \beta$$

$$\beta \text{ LUB} + \alpha \text{ UB} \Rightarrow \beta \leq \alpha$$

$$\Rightarrow \alpha = \beta$$



Convention If $A = \emptyset$, then any element is an UB for A .

$\Rightarrow$ LUB DNE

GLB DNE

Now we assume

(P13) LUB Property

If A is a subset of system and $A \neq \emptyset$, A is bounded above then LUB of A exists in the system.

Note that $\mathbb{P}$ fails for $\mathbb{Q}$

$$A := \{x < \sqrt{2}\} \cap \mathbb{Q}$$

$$= \left\{ \begin{array}{c} x < 0 \\ \text{in } \mathbb{Q} \end{array} \right\} \cup \left\{ \begin{array}{c} x^2 < 2 \\ \text{in } \mathbb{Q} \end{array} \right\}$$

LUB should be $\sqrt{2}$, but $\sqrt{2} \notin \mathbb{Q}$, so it DNE in $\mathbb{Q}$.

Proof of IVT wolog fcts. on $[a, b]$

$f(a) < 0 < f(b)$. Then $\exists x \in [a, b]$ s.t. $f(x) = 0$

Pf. Define $A := \{x \in [a, b] : f(y) \leq 0 \ \forall y \in [a, x]\}$ $\rightarrow$ want to apply P13.

- $A \neq \emptyset$: Since $f(a) < 0$, so $a \in A$, $a \neq \emptyset$
- A is bounded above

b is an upper bound by construction

P13

$\Rightarrow$ We claim that the α in the statement of IVT is $\sup A$.

Suppose for a contradiction that $f(\alpha) \neq 0$

Case 1 $f(\alpha) > 0$ (P10)

Since positivity of a continuous function is an open property,

$\exists \delta > 0$ s.t. $f(x) > 0$ for $\forall x \in (\alpha - \delta, \alpha + \delta)$.

Then $f(\alpha - \frac{\delta}{2}) > 0$

so $\alpha - \frac{\delta}{2}$ is an upper bound for A .

Since $\alpha - \frac{\delta}{2}$ is a smaller UB than α , this contradicts the minimality of α .

Case 2 $f(\alpha) < 0$

Since negativity of a continuous function is an open property, $\exists \delta > 0$ s.t. $\forall x \in (\alpha - \delta, \alpha + \delta)$, $f(x) < 0$

Since $f(x) < 0$ on $[a, \alpha)$, $f(x) < 0$ on $(\alpha, \alpha + \delta]$ $\Rightarrow \alpha + \frac{\delta}{2} \in A$.

So α is not even an UB, let alone the UB, contradiction.

So $f(\alpha) = 0$ by Trichotomy $\Rightarrow$ IVT is proven



Pf of Boundedness Thm

f cts. on $[a, b]$, then f bounded above.

Pf. Let $A := \{x \in [a, b] : f \text{ bounded above } [a, x]\}$

- $A \neq \emptyset$ b/c. $a \in A$
- A is bounded above b/c. by construction, b is the largest element that could be in A .

So by P13, let α be the LUB of A .

(want $\alpha = b$ at the very least but then we have to check that we don't have)